

Total No. of Questions : 10]

SEAT No. :

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[Total No. of Pages :2

T. E. (Computer Engineering)

DIGITAL SIGNAL PROCESSING APPLICATIONS

(2012 Course) (Semester-II) (End Sem.) (310253)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

Q1) a) State the sampling theorem and explain in brief the coding process of ADC. [5]

b) Obtain the Z Transform(ZT) of a DT signal

$x(n) = a^n u(-n-1)$ Sketch the ROC. [5]

OR

Q2) a) A CT signal having frequency 50 Hz is sampled at a rate of 1200 samples /sec. Obtain

- i) Number of samples per cycle.
- ii) Digital/Discrete frequency f and ω .
- iii) Minimum sampling rate to avoid aliasing effect.
- iv) Period of a DT signal. [5]

b) State and prove the time shifting property of Fourier Transform(FT). Define it for DFT. [5]

Q3) a) Compare Linear Convolution with Circular Convolution. [5]

b) Define N point DFT by means of twiddle factor and obtain the twiddle factors for 4 point DFT. [5]

OR

Q4) a) Obtain the computational complexity of Radix-2 DIF FFT Algorithm. [5]

b) Define the system, function and obtain it for the given system described

as- $y(n) - \frac{5}{6} y(n-1) + \frac{1}{6} y(n-2) = x(n) - \frac{1}{2} x(n-1)$ [5]

P.T.O.

Q5) a) Derive the Direct Form-II IIR filter structure from the system function $H(Z)$ and realize it using multipliers, adders and delay elements. [9]

b) Obtain and realize linear phase FIR filter structure having impulse response

$$h(n) = \delta(n) + \frac{1}{2}\delta(n-1) - \frac{1}{4}\delta(n-2) + \frac{1}{2}\delta(n-3) + \delta(n-4) \quad [9]$$

OR

Q6) a) Obtain and draw the cascade form realization for IIR filter having transfer

function
$$H(Z) = \frac{5Z(Z + 0.4)}{(Z - 0.2)(Z - 0.6)} \quad [9]$$

b) Represent the mathematical form of Nth order FIR filter by means of system function $H(Z)$. Draw the Direct Form filter structure and determine the number of multipliers, adders and delay elements required to realize the filter. [9]

Q7) a) Compare conventional Microprocessor with DSP Processor architecture. Draw and explain basic building blocks of DSP processor. [8]

b) Draw and explain the SIMD(Single Instruction Multiple Data) architecture of SHARC DSP processor. [8]

OR

Q8) a) Explain the features of SHARC DSP processor. List the number of DAGs with its capabilities and memory pointer registers supported by DAG. [8]

b) What is OMAP? Explain the features and applications of OMAP in brief. [8]

Q9) a) What is Companding? What is its significance in audio processing? What is the impact of data rate on sound quality? [8]

b) With mathematical form, explain any two gray level transforms used for image enhancement. [8]

OR

Q10) a) Draw and explain Human Speech Model in speech synthesis and recognition. [8]

b) How digital image is represented by means of digital computer? How gray scale image is different than colour image? What is Histogram of an image? [8]

